

Title: Uncovering patterns in neural circuits and their link with cellular identity

Description: Our ability to consolidate every-day experiences into lasting memories is a fundamental cognitive process, yet the precise neural circuits that govern this transformation remain largely uncharted. This project aims to discover the underlying complex identity and interactions of neurons that mediate memory processing.

Data are available through a powerful combination of cutting-edge techniques to investigate the relationship between a neuron's genetic identity, its anatomical properties and input-output partners. Advanced methods have been used to trace the long-range and local projections of individual neurons and collect information on large-scale gene expression. By integrating this data, we aim to classify neurons into distinct subtypes and understand their specific roles in memory processing and inter-areal communications.

The project will utilize different multimodal datasets to bridge the gap between cellular identity and circuit connectivity. Students will be provided with ~170 single-neuron axonal reconstructions (see Figure 1), with the goal of understanding the main patterns in connectivity and their relation with cell-body position. This will be supplemented by large-scale MERFISH datasets (Zhang et al. 2023) for broader molecular context. The primary objective is to integrate these projectomic and transcriptomic datasets, to understand the link between connectivity and genetic makeup.

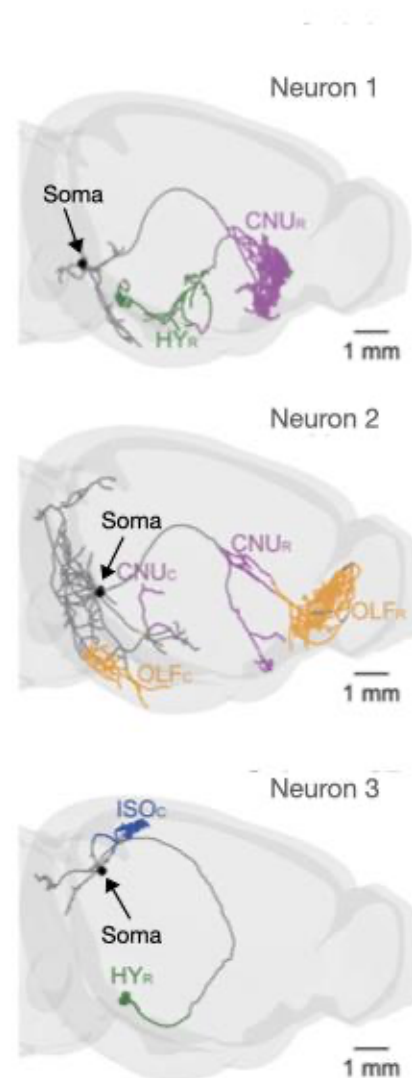


Figure 1: Examples of single neuron reconstruction for three neurons (Qui et al. 2024). The cell body (soma) is marked with a black dot, and single-neuron reconstruction traces the axonal and branching structure for each neuron characterizing its projection to other brain areas. Branches terminating in specified target areas are shown in color (abbreviations in Table S3, Qui et al. 2024).

Questions: The project aims to answer the following questions:

1. What are the main patterns in connectivity? Do neurons target only a single region? Or do they branch out more widely to many brain areas? For this, you will employ clustering methods to uncover the main patterns in connectivity in single-neuron axonal reconstruction data.
2. How does the cell-body position of the neuron influence its projections? For this, you will employ classification methods to link the identified clusters with cell-body coordinates.
3. How does the cell-body position of the neuron relate to its gene expression? For this, you will employ supervised learning methods to link gene expression with cell-body coordinates.
4. How does genetic makeup influence brain connectivity? For this, you will link the datasets through their cell-body coordinates, which have already been mapped to a common reference.

For ambitious students, spatial transcriptomic data from the Surlin lab—generated via BARseq (Yuan et al. 2024) provides expression profiles for 64 key genes (including cadherins and catenins) across the mouse hippocampal formation. This further complements the analysis by providing both transcriptomic and projectomic data for thousands of cells. This data is collected using advanced molecular imaging techniques, requiring various processing steps before analysis.

References:

- Qiu, S., *et al.* "Whole-brain spatial organization of hippocampal single-neuron projectomes". *Science* 383: 6682 (2024). DOI:10.1126/science.adj9198
- Yuan, Li, Xiaoyin Chen, Huiqing Zhan, Gilbert L. Henry, and Anthony M. Zador. 2024. "Massive Multiplexing of Spatially Resolved Single Neuron Projections with Axonal BARseq." *Nature Communications* 15 (1): 8371.
- Zhang, Meng, Xingjie Pan, Won Jung, et al. 2023. "Molecularly Defined and Spatially Resolved Cell Atlas of the Whole Mouse Brain." *Nature* 624 (7991): 343–354.